Part 1 - Upper Bound

Let S be a non empty set of real numbers.

Suppose there is a real number B such that

$$B \geq x, \quad \forall x \in S$$

Then B is the upper bound for S.

meam:- B is greater than or equal to every element of S.

Ex:- $S = \{1, 2, 3, 4\}$

We check: $5 \geq 1$, $5 \geq 2$, $5 \geq 3$, $5 \geq 4$

All are True so 5 is an upper bound of S.

4 is also an upperbound.

There can be many upper bounds

An upper bound does not have to belong to the set.

Take $S = (0, 4)$

Here, 5 is an upper bound.

But, $5 \notin S$. That is okay.

Bounded Above

- If a set has least one upper bound, we say:

The set is bounded above.

For ex:- $S = (0, 4)$ is bounded above because 4, 5, 6 etc are upper bounds.

unbounded set

- If there is no number whatsover that can be an upper bound, then the set is unbounded above.

Ex: $S = \{1, 2, 3, 4, 5 \ldots\}$

The set of all positive integers.

Suppose we say 100 is UB.

But $101 \in S$ and $101 > 100$

So, 100 isn't an UB

What about 1,000,000? - There is always a larger positive integer

Therefore: Positive integers are unbounded above.

Part II - SUPREMUM

- The smallest upperbound is called the supremum. or **least upper bound.**

Formal Definition :-

A real number B is called the supremum of S if:

Condition 1

B is an upper bound.

that means:

$$B \geq x, \quad \forall x \in S$$

Condition 2

No number smaller than B is an upper bound.

Ex:- $S = (0, 4]$ $\quad \nexists B'$ such that $B' < B$ (Another UB)

lets prove $\sup S = 4$

Step 1 : Is 4 an upper bound? Yes

Every $x \in S$ satisfies $x \leq 4$

Therefore, 4 is an UB.

Step 2: Is there a smaller UB?

Suppose we try 3.9

But $4 \in S$

and $4 > 3.9$

Therefore, 3.9 is not UB

$\therefore \sup S = 4$

SUPREMUM vs MAXIMUM

- The supremum may or may not belong to the set. If the supremum

belongs to the set, it is called the maximum.

Case 1: $S = [0,4]$

We know, $\sup S = 4$

Is $4 \in S$?

Yes.

Therefore: $\max S = 4$

So Here: $\sup S = \max S = 4$

Case 2: $S = [0,4)$

We know, $\sup S = 4$

But, $4 \notin S$

Therefore, there is no maximum

So, $\sup S = 4$ but no maximum exists

> Supremum = smallest UB
>
> Maximum = largest element actually inside the set.

Part III - ε-Definition of Supremum

Let S be the non empty set of $\mathbb{R}$, the suprema $(S, \leq)$ B will be suprema of S if:-

① B is the UB

② $\forall \varepsilon > 0 \quad \exists x \in S$ if $x \geq B - \varepsilon$

↳ strict inequality or weak inequality

Lower Bound

Suppose $S = \{1, 2, 3, 4\}$

A number L is LB if

$L \leq x \quad \forall x \in S$

For ex: $0, -1, -100, 1$ are all LB as the elements of S are greater than or equal to those numbers.

Greatest Lower Bound - INFIMUM

Formally, L is the infimum if:

Condition 1

$$L \leq x \quad \forall x \in S$$

So, L is a lower bound.

Condition 2.

No number greater than L is a LB.

Ex: $S = [0, 4]$

$0 \in S$ still $\inf S = 0$

Because $0 \leq x$ for every $x \in S$.

But any number greater than 0 say 0.1 isn't in the set.

INFIMUM VS MINIMUM

Infimum need not belong to the set.

But Minimum must belong to the set.

If $\inf S \in S$

Then, $\min S = \inf S$

Axiom ⇒ Statement that is true but cannot be proven

S: Non empty and real

↳ BA ⇒ It has suprema

Let A and B be two sets.

A, B ⊆ ℝ and nonempty and Bounded Above.

Let sup(A) = α

sup(B) = β

Claim :: Define A+B = {x+y : x ∈ A, y ∈ B}

Claim: sup(A+B) = sup(A) + sup(B)

Some examples:

A = [1,4], B = {0.7}

A+B = ?

Ans ⇒ [1.7, 4.7]

A = {1,4}, B = {0.7}

A+B = {1.7, 4.7}

A = [1,4), B = {0.7, 0.9}

A+B = [1.7, 4)

A = [1,4)

B = {0.7}

sup A = 4

sup B = 0.7

A+B = [1.7, 4.7]

sup(A+B) = sup(A) + sup(B)

4.7 = 4 + 0.7

Show that α+β is a suprema of A+B

Given,

sup A = α ⇒ x ≤ α ∀ x ∈ A

$\sup(B) = \beta \Rightarrow y \leq \beta \quad \forall y \in \beta$

$x + y \leq \alpha + \beta \quad \forall x \in A \; \& \; y \in B$

$\alpha + \beta$ is the upper bound of the set $A + B$.

Is $\alpha + \beta$ a least UB?

$\sup(A) = \forall \varepsilon' > 0 \; \exists x \in A \; s.t \; x > \alpha - \varepsilon'$

$\sup(B) = \forall \varepsilon'' > 0 \; \exists y \in B \; st \; y > \beta - \varepsilon''$

By adding them,

$\forall \varepsilon_1 + \varepsilon'' > 0 \quad \exists$ some $x + y > \alpha + \beta - \varepsilon$